

ROSHAN BALLAV RAUT

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EDUCATION

A. P. Shah Institute of Technology — Kalyan/Thane, India

B.E., Computer Engineering (AI & ML) · Mumbai University · 2025 – 2029

Pre-College Scores: SSC 84.6% | HSC 71.33% | MHT-CET 87.9 Percentile

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL

Web / Backend: Spring Boot, Flask, REST APIs, LangChain4j, HTML, CSS, Jinja2

Databases: PostgreSQL, pgvector (vector search), SQLite

AI / ML: PyTorch (foundations), Agentic RAG Architecture, Embeddings, Prompt Engineering, LLM APIs (Gemini, Groq)

APIs & Tools: Google GenAI SDK (Gemini), Groq API, Alpha Vantage API, Git, GitHub, pytest

Core CS: DSA, Search & Sort Algorithms, Memory Management (Pointers, C), OOP, HTTP Request-Response Cycle, Regular Expressions

PROJECTS

ForgeRAG – Industrial-Grade Agentic RAG System

Academic Mini-Project · In Progress

Java 17 · Spring Boot · LangChain4j / LangGraph4j · PostgreSQL (pgvector) · Gemini API

- Building a stateful, self-correcting Agentic RAG backend modeled on enterprise compliance engines: a LangGraph4j state machine routes between Retrieval, Generation, and Evaluation nodes, using an LLM-as-judge to catch hallucinations and trigger a query-rewrite retry loop before the final answer.
- Using structure-aware/semantic chunking to preserve document hierarchy and normalized-embedding vector search via pgvector Inner Product indexing, targeting multi-hop reasoning and compliance/audit use cases.

Extractly – RAG-Based Document Q&A Web App

CS50x Final Project

Flask · PostgreSQL (pgvector) · Gemini Embeddings/Generation · pypdf · PyMuPDF · Gemini Vision

- Built a full-stack RAG Q&A platform: PDF/image ingestion, OCR fallback via PyMuPDF and Gemini Vision, vector embeddings stored in PostgreSQL with pgvector, and Gemini-generated answers grounded in retrieved context.
- Shipped end-to-end and resolved real production issues – deprecated SDK migration, Supabase IPv6/session-pooler constraints, and an auth refactor from JSON/fetch to form-POST with flash messaging.

FinHabits – AI-Powered Personal Finance Parser

Built for Google Solution Challenge 2026

Python · React · Groq (Llama 3) · Gemini

- Designed a 3-layer inference pipeline (regex → Groq Llama 3 → Gemini fallback) to parse and categorize Indian bank SMS transactions without manual entry.

CrisisSync – AI Emergency Coordination Platform

Proposed for Google Solution Challenge 2026 · GDG on Campus India (Concept)

Flutter · Gemini AI · Firebase · Google Maps (planned)

- Designed and prototyped the front-end UI/UX for an AI hotel emergency-response concept: guest SOS flow, command-center dashboard, and floor-level incident map, with a Gemini-powered triage engine and modified-Dijkstra evacuation-routing approach as the design basis for future backend work.

COURSEWORK & CURRENTLY EXPLORING

- CS50x: Introduction to Computer Science – Harvard University · Final project (Extractly) complete
- CS50P: Intro to Programming with Python – Harvard University · Final project (Flashcard Quiz CLI) in progress
- PyTorch Foundations – hands-on session building and training a circle classifier; deepening into advanced retrieval/embedding systems and production AI serving
- CS50AI (Python) – still exploring/watching lectures on neural networks & deep learning fundamentals
- Still learning Java DSA via Striver's A2Z Sheet (A2Z_DSA_Java repo, per-problem solutions + notes)
- Studying networking/systems fundamentals for better AI system design – latency, DNS, TCP, TLS, IP addressing